

BIOS 735: Shrinkage Regression Methods

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“There's two
buttons I
never like
to hit: that's
panic and
snooze”

Introduction

- ▶ Today, we'll talk about shrinkage methods
- ▶ All shrinkage methods are biased (by design) even when the model is correct.
- ▶ They will, however, result in lower variance than the full LR model.
- ▶ Some shrinkage methods we'll discuss:
 - ▶ Ridge Regression
 - ▶ Lasso
 - ▶ Elastic Net
 - ▶ Least Angle Regression (LAR)

Instability of LS Estimators

- ▶ Recall that $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ can be unstable
- ▶ Further, when $\text{rank}(\mathbf{X}) < p$ $\mathbf{X}'\mathbf{X}$ is singular
 - ▶ $\mathbf{X}$ is ill-conditioned
 - ▶ columns of $\mathbf{X}$ are collinear
 - ▶ $p > n$
- ▶ We can measure ill-conditioning by $\kappa = d_1/d_p$ where d_1 and d_p are the largest and smallest eigenvalues, respectively, of $\mathbf{X}$.
- ▶ We can measure collinearity for a predictor X_k via $VIF_k = (1 - R_k^2)^{-1}$ where R_k^2 is the R^2 for all other covariates regressed on X_k

Biased Regression Methods

- ▶ The instability of the LS estimates is due to $\mathbf{X}'\mathbf{X}$ being singular or nearly singular.
- ▶ **Solution** allow $\hat{\beta}$ to be biased.
- ▶ We'll assume that $\mathbf{x}$ and $\mathbf{y}$ have been centered (no β_0). In practice, we will center and scale these data.

Ridge Regression

- ▶ Potential instability in $(\mathbf{X}'\mathbf{X})^{-1}$ could be solved by adding a constant λ to $\mathbf{X}'\mathbf{X}$.
- ▶ Adding a constant to the diagonal of $\mathbf{X}'\mathbf{X}$ will bring it closer to a matrix with 0's on the off diagonal.
- ▶ This results in the ridge regression estimate of $\hat{\beta}$

$$\hat{\beta}^{\text{ridge}} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}'\mathbf{y}$$

where $\mathbf{I}$ is the $p \times p$ identity matrix.

Ridge Regression Properties

- ▶ Another way to view the ridge regression model is that

$$\hat{\beta}^{\text{ridge}} = \operatorname{argmin}_{\beta} \{ (\mathbf{y} - \mathbf{x}\beta)'(\mathbf{y} - \mathbf{x}\beta) + \lambda \|\beta\|^2 \},$$

where $\|\beta\|^2 = \sum_{j=1}^p \beta_j^2$ (note: the λ here is the same as on the previous slide)

- ▶ (Yet) Another way to view the ridge regression model is that $\hat{\beta}^{\text{ridge}}$ minimizes

$$ESS(\beta) = (\mathbf{y} - \mathbf{x}\beta)'(\mathbf{y} - \mathbf{x}\beta) \quad \text{such that} \quad \|\beta\|^2 \leq t,$$

which makes explicit the size constraint on the parameters.

- ▶ The relationship between t and λ is made through Lagrange multipliers.

Ridge Regression Properties

- The bias and variance of Ridge coefficients have nice closed forms

$$\text{Var}_\lambda(\hat{\beta}^{\text{ridge}}) = \sigma^2 \sum_{j=1}^p \frac{d_j^2}{(d_j^2 + \lambda)^2}$$

$$\text{Bias}_\lambda^2(\hat{\beta}^{\text{ridge}}) = \lambda^2 \sum_{j=1}^p \frac{(\hat{\alpha}_j^{\text{ridge}})^2}{(d_j^2 + \lambda)^2}$$

thus,

$$\text{MSE}_\lambda(\hat{\beta}^{\text{ridge}}) = \sigma^2 \sum_{j=1}^p \frac{\sigma^2 d_j^2 + \lambda (\hat{\alpha}_j^{\text{ridge}})^2}{(d_j^2 + \lambda)^2}$$

Lasso regression model

- ▶ The Lasso regression model, results in solution $\hat{\beta}^{\text{lasso}}$ minimizes

$$ESS(\beta) = (\mathbf{y} - \mathbf{x}\beta)'(\mathbf{y} - \mathbf{x}\beta) \quad \text{such that} \quad \sum_{i=1}^p |\beta_j| \leq t,$$

which makes explicit the size constraint on the parameters.

- ▶ The solution to this is such that

$$\hat{\beta}^{\text{lasso}} = \operatorname{argmin}_{\beta} \left\{ (\mathbf{y} - \mathbf{x}\beta)'(\mathbf{y} - \mathbf{x}\beta) + \lambda \sum_{i=1}^p |\beta_j| \right\},$$

- ▶ The relationship between these forms is made through Lagrange multipliers.

Lasso properties

- ▶ A major difference between the Lasso and Ridge models is that making t sufficiently small (or λ large) will result in some β coefficients being exactly zero.
- ▶ Let $t_0 = \sum_{j=1}^p |\hat{\beta}_j^{ls}|$ then if $t = t_0/2$ the Lasso will result in coefficients that are the least squares coefficients shrunk by about 50% on average.
- ▶ There is no closed-form solution for Lasso coefficients (they are nonlinear in $\mathbf{y}$).

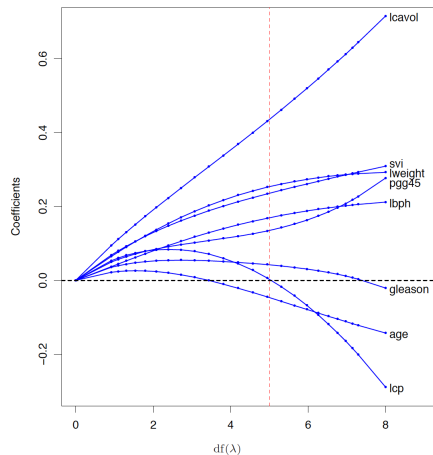
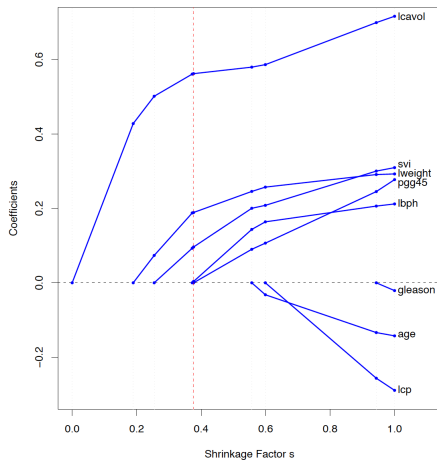


Figure: From ESL, where $s = t / \sum_1^p |\hat{\beta}_j^{ls}|$

Ridge vs Lasso vs Elastic Net

- ▶ The Lasso regression model,

$$\hat{\beta}^{\text{lasso}} = \operatorname{argmin}_{\beta} \left\{ (\mathbf{y} - \mathbf{x}\beta)'(\mathbf{y} - \mathbf{x}\beta) + \lambda \sum_{i=1}^p |\beta_j| \right\},$$

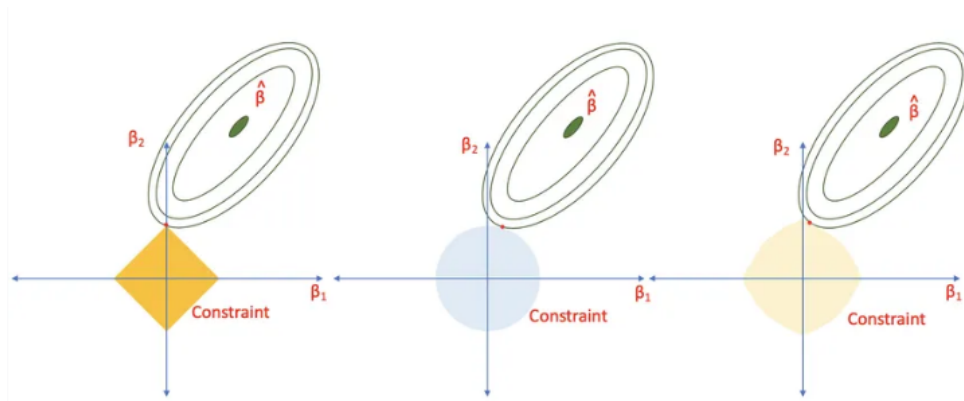
- ▶ The ridge regression model,

$$\hat{\beta}^{\text{ridge}} = \operatorname{argmin}_{\beta} \left\{ (\mathbf{y} - \mathbf{x}\beta)'(\mathbf{y} - \mathbf{x}\beta) + \lambda \sum_{i=1}^p \beta_j^2 \right\},$$

- ▶ The elastic net regression model (Zou and Hastie, 2005),

$$\hat{\beta}^{\text{en}} = \operatorname{argmin}_{\beta} \left[(\mathbf{y} - \mathbf{x}\beta)'(\mathbf{y} - \mathbf{x}\beta) + \lambda \sum_{i=1}^p \{ \alpha \beta_j^2 + (1 - \alpha) |\beta_j| \} \right],$$

Ridge vs Lasso vs Elastic Net



Ridge vs Lasso vs Best Subset

- Suppose $\mathbf{X}$ is a standardized orthonormal input matrix
- The estimators for Ridge, Lasso, and best subset, denoted by $\hat{\beta}^{\text{ridge}}$, $\hat{\beta}^{\text{lasso}}$ and $\hat{\beta}^{\text{bss}}$ respectively, are

$$\begin{aligned}\hat{\beta}_j^{\text{bss}} &= \hat{\beta}_j^{\text{ls}} \times I\left(|\hat{\beta}_j^{\text{ls}}| \geq |\hat{\beta}_{(M)}|\right) \text{ of size } M \\ \hat{\beta}_j^{\text{ridge}} &= \frac{\hat{\beta}_j^{\text{ls}}}{1 + \lambda} \\ \hat{\beta}_j^{\text{lasso}} &= \text{sign}\left(\hat{\beta}_j^{\text{ls}}\right) \left(|\hat{\beta}_j^{\text{ls}}| - \lambda\right)_+\end{aligned}$$

where $\hat{\beta}_{(M)}$ is the M th largest regression coefficient.

Other penalties

- ▶ Last class, we went over Ridge, LASSO, and elastic net regression models.
- ▶ The optimization functions were all of the form

$$\operatorname{argmin}_{\beta} \{ (\mathbf{y} - \mathbf{x}\beta)'(\mathbf{y} - \mathbf{x}\beta) + p(\beta; \lambda) \},$$

where $p(\beta; \lambda)$ is a penalty function.

- ▶ There are other options for $p(\beta; \lambda)$.

Minimax Concave Penalty (MCP)

- ▶ Given a tuning parameter $\lambda > 0$ and a parameter $a > 1$, the MCP penalty for a coefficient β is defined as:

$$p_{MCP}(\beta; \lambda, a) = \begin{cases} \lambda|\beta| - \frac{\beta^2}{2a} & \text{if } |\beta| \leq a\lambda \\ \frac{a\lambda^2}{2} & \text{if } |\beta| > a\lambda \end{cases}$$

- ▶ The first segment is concave, and the penalty becomes constant after $|\beta| > a\lambda$.
- ▶ The parameter a controls the concavity and thus influences the amount of regularization applied to the coefficients.

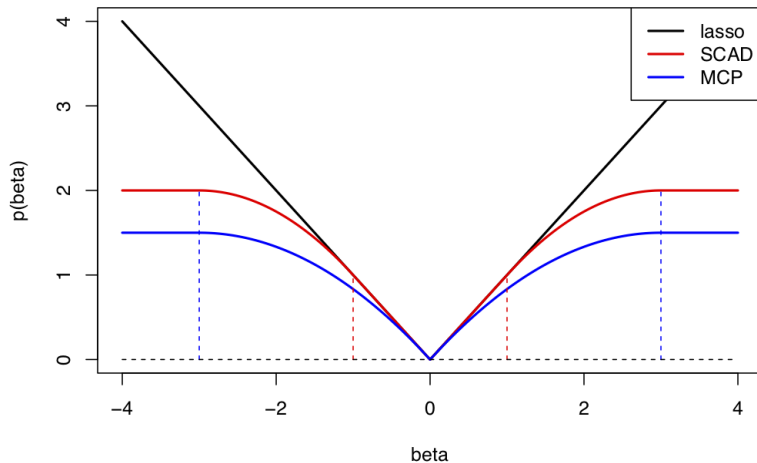
Smoothly Clipped Absolute Deviation (SCAD)

- ▶ Given a tuning parameter $\lambda > 0$ and a parameter $a > 2$, the SCAD penalty for a coefficient β is defined as:

$$p_{SCAD}(\beta; \lambda, a) = \begin{cases} \lambda|\beta| & \text{if } |\beta| \leq \lambda \\ -\left(\frac{\beta^2 - 2a\lambda|\beta| + \lambda^2}{2(a-1)}\right) & \text{if } \lambda < |\beta| \leq a\lambda \\ \frac{(a+1)\lambda^2}{2} & \text{if } |\beta| > a\lambda \end{cases}$$

- ▶ The SCAD penalty starts linearly (similar to the L1 penalty) but then bends down and becomes flat after $|\beta| > a\lambda$.
- ▶ The SCAD penalty is continuous everywhere, making it smoother than the MCP.

Lasso, SCAD and MCP penalties



Partial Least Squares

- ▶ PCR was performed by making a linear combination of the $\mathbf{X}$'s, then performing LR on the derived linear combinations.
- ▶ Partial Least Squares (PLS) has similarities with Principal Components Analysis (PCA), except that the linear combination of the $\mathbf{X}$'s is made using the outcome.

Partial Least Squares Algorithm

1. Standardize each $\mathbf{x}_j$ and set $\hat{\mathbf{y}}^{(0)} = \mathbf{y} - \bar{\mathbf{y}}\mathbf{1}$, and $\mathbf{x}_j^{(0)} = \mathbf{x}_j$, $j = 1, 2, \dots, p$.
2. For $m = 1, 2, \dots, M \leq p$
 - 2.1 $z_{im} = \sum_{j=1}^p \hat{\psi}_{mj} x_{ij}^{(m-1)}$, where $\hat{\psi}_{mj}$ is the SLR coefficient for $\mathbf{x}_j^{(m-1)}$ on $\mathbf{y}^{(m-1)}$

$$\hat{\psi}_{mj} = \frac{\mathbf{x}_j^{(m-1)'} \mathbf{y}}{\mathbf{x}_j^{(m-1)'} \mathbf{x}_j^{(m-1)}} = \frac{\langle \mathbf{x}_j^{(m-1)}, \mathbf{y} \rangle}{\langle \mathbf{x}_j^{(m-1)}, \mathbf{x}_j^{(m-1)} \rangle}$$

- 2.2 Get the SLR coefficient for $\mathbf{z}_m$ on $\mathbf{y}^{(m-1)} \Rightarrow \hat{\theta}_m = \langle \mathbf{z}_m, \mathbf{y}^{(m-1)} \rangle / \langle \mathbf{z}_m, \mathbf{z}_m \rangle$
 - 2.3 $\hat{\mathbf{y}}^{(m)} = \hat{\mathbf{y}}^{(m-1)} - \hat{\theta}_m \mathbf{z}_m$.
 - 2.4 Orthogonalize each $\mathbf{x}_j^{(m-1)}$ with respect to $\mathbf{z}_m \Rightarrow \mathbf{x}_j^{(m)}$
3. Then the PLSR model with m components is

$$\hat{\mathbf{y}}(k) = \bar{\mathbf{y}}\mathbf{1} + \sum_{m=1}^k \hat{\theta}_m \mathbf{z}_m.$$

Grouped Lasso regression model

- ▶ In some problems, the predictors belong to predefined groups (e.g., genes belonging to a specific biological pathway or dummy-coded categorical variables).
- ▶ In this situation, it may be desirable to shrink and select the members of a group together.
- ▶ Suppose that the p predictors are divided into L groups, with p_ℓ being the number of variables in group ℓ
- ▶ The Grouped Lasso regression model minimizes

$$(\mathbf{y} - \mathbf{x}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{x}\boldsymbol{\beta}) + \lambda \sum_{\ell=1}^L \sqrt{p_\ell} \|\boldsymbol{\beta}_\ell\|_2,$$

where $\boldsymbol{\beta}_\ell$ are the regression coefficients for group ℓ , and $\|\cdot\|_2$ is the Euclidean norm.

Grouped Example: Genetic variants grouped by genes

Suppose we want to predict disease risk using thousands of SNPs.

- ▶ SNPs can be naturally grouped according to the **gene** in which they occur.
- ▶ Rather than selecting individual SNPs independently, the grouped lasso can select or remove **entire genes**.
- ▶ This incorporates known biological structure into the variable-selection procedure.

$$\underbrace{\beta_1, \dots, \beta_5}_{\text{Gene 1}} \quad \underbrace{\beta_6, \dots, \beta_9}_{\text{Gene 2}} \quad \underbrace{\beta_{10}, \dots, \beta_{15}}_{\text{Gene 3}}$$

Why grouped lasso?

The scientific question may be whether a **gene or biological pathway** is predictive, rather than whether one particular SNP is predictive.

Fused Lasso regression model

- ▶ Along with grouped methods, there is the idea of fusion or merging.
- ▶ In this situation, it may be desirable to shrink and select the members of a group towards the same regression coefficient.
- ▶ Consider the categorical covariates; by assuming the coefficient for one group is zero, you fuse that group with the referent group.
- ▶ The idea of fusion is to fuse groups together and not have the fusion be subject to the reference group.
- ▶ The Fused Lasso regression proposed by Tibshirani et al (2005) minimizes

$$\hat{\beta} = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) + \lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=2}^p |\beta_j - \beta_{j-1}|$$

Fused Example: Detecting regions of DNA with copy number alterations

For each patient, measurements are collected at thousands of **ordered locations along a chromosome**.

- ▶ Nearby genomic locations tend to have similar copy numbers.
- ▶ A deletion or amplification generally affects a **contiguous region** of the chromosome.
- ▶ The fused lasso penalizes differences between coefficients at neighboring locations:

$$\lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=2}^p |\beta_j - \beta_{j-1}|$$

Why fused lasso?

The fusion penalty encourages neighboring locations to have the same estimated effect, producing **piecewise-constant genomic regions**.